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Phase Contrast X-Ray Imaging Apparatus, X-Ray Image Processing Apparatus, X-Ray Image Processing Method and Correction Curve Generation Method

Abstract

A phase contrast X-ray imaging apparatus (**100**) according to this disclosure includes an X-ray source (**1**); an X-ray detector (**2**); a plurality of gratings; an image processor (**7***a*) for generating a first dark field image (**35***a*); a storage (**8**) for storing a plurality of correction curves (**20***a*) generated at positions in a direction perpendicular to a grating direction; and a controller (**7***b*) for correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of a subject (**90**) and values that relate to X-ray scattering of the subject at corresponding one of the positions.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a phase contrast X-ray imaging apparatus, an X-ray image processing apparatus, an X-ray image processing method and a correction curve generation method, and in particular to a phase contrast X-ray imaging apparatus, an X-ray image processing apparatus, an X-ray image processing method and a correction curve generation method using a plurality of gratings.

BACKGROUND ART

[0002] Phase contrast X-ray imaging apparatuses using a plurality of gratings for capturing images are known in the art. Such a phase contrast X-ray imaging apparatus is disclosed in Japanese Patent Laid-Open Publication No. JP 2021-194389, for example.

[0003] Japanese Patent Laid-Open Publication No. JP 2021-194389 discloses a radiography system including a radiation generator, a radiation detector, a plurality of gratings arranged between the radiation generator and the radiation detector, and an image processing apparatus for generating a small angle scattering image based on signals detected by the radiation detector.

PRIOR ART

Patent Document

[0004] Patent Document 1: Japanese Patent Laid-Open Publication No. JP 2021-194389

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

[0005] Here, although not stated in Japanese Patent Laid-Open Publication No. JP 2021-194389, it is known that, when radiation (X-rays) passes through an object, a spectrum of X-rays changes. A hardening degree of the spectrum of X-rays varies in accordance with a thickness of the object through which the X-rays pass (a path length of X-rays). The change of the spectrum of X-rays also occurs when X-rays pass through the grating.

[0006] In radiography systems (phase contrast X-ray imaging apparatuses) such as the radiography system disclosed in Japanese Patent Laid-Open Publication No. JP 2021-194389, X-rays that radiate from a radiation generator (an X-ray source) radiate in a so-called cone-beam form. In this case, an incident angle of X-rays in a peripheral part of a grating is larger than a central part of the grating. If incident angles of X-rays with respect to the grating are different, path lengths of the X-rays passing through the grating correspondingly changes so that change degrees of spectra of X-rays become different. For this reason, difference between change degrees of spectra of X-rays causes uneven noise in a small angle scattering image (dark field image). In this case, there is a problem that image quality of the dark field image decreases due to noise caused by X-rays incident on the grating in an inclined direction.

[0007] The present invention is intended to solve the above problem, and one object of the present invention is to provide a phase contrast X-ray imaging apparatus, an X-ray image processing apparatus, an X-ray image processing method and a correction curve generation method capable of reducing deterioration of image quality of a dark field image even in a case in which X-rays are incident on a grating in an inclined direction.

Means for Solving the Problems

[0008] In order to attain the aforementioned object, the present inventors have diligently studied uneven noise, and as a result have found that uneven noise appears in a dark field image due to change of a scattering degree of X-rays caused by interaction between change of a spectrum of X-rays appearing in transmission of the X-rays through a subject and change of a spectrum of X-rays caused by difference between incident angles of the X-rays on a grating. That is, a phase contrast X-ray imaging apparatus according to a first aspect of the present invention includes an X-ray source; an X-ray detector for detecting the X-rays for irradiation from the X-ray source; a plurality of gratings arranged between the X-ray source and the X-ray detector; an image processor for generating a phase contrast X-ray image including a first dark field image, which is a dark field image of a subject, based on an intensity distribution of the X-rays detected by the X-ray detector; a storage for storing a plurality of correction curves generated at positions in a direction perpendicular to a grating direction in which the plurality of gratings extend; and a controller for correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions.
[0009] An X-ray image processing apparatus according to a second aspect of the present invention includes an image acquirer for acquiring a phase contrast X-ray image including a first dark field image, which is a dark field image of a subject; a storage for storing a plurality of correction curves generated at positions in a direction perpendicular to a grating direction in which a plurality of gratings extend; and a controller for correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions.
[0010] An X-ray image processing method according to a third aspect of the present invention includes acquiring a phase contrast X-ray image including a first dark field image, which is a dark field image of a subject; storing a plurality of correction curves generated at positions in a direction perpendicular to a grating direction in which a plurality of gratings extend; and correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions.
[0011] A correction curve generation method according to a fourth aspect of the present invention includes acquiring a second dark field image, which is a dark field image of a correction phantom, by capturing an image of the correction phantom by using a phase contrast X-ray imaging apparatus including a plurality of gratings; acquiring a second absorption image, which is an absorption image of the correction phantom; and generating the plurality of correction curves at positions in the direction perpendicular to the grating direction in which the plurality of gratings extend by using the second dark field image and the second absorption image.
Effect of the Invention
[0012] In the phase contrast X-ray imaging apparatus according to the first aspect, the X-ray image processing apparatus according to the second aspect, and the X-ray image processing method according to the third aspect, a plurality of correction curves corresponding to the positions in a direction perpendicular to the grating direction is stored. Because each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions, a change degree of a spectrum of X-rays depending on an incident angle of the X-rays at each position is taken into account. Also, the first dark field image can be accurately corrected by correcting the first dark field image, which is a dark field image of a subject, by using the correction curves corresponding to the positions, as compared with a case in which the first dark

field image is corrected by using a single common correction curve. Consequently, it is possible to reduce deterioration of image quality of a first dark field image even in a case in which X-rays are incident on a grating in an inclined direction.
[0013] In the correction curve generation method according to the fourth aspect of the present invention, because the plurality of correction curves are generated by using the second dark field image and the second absorption image acquired by actually capturing the correction phantom, it is possible to generate the plurality of correction curves with a change degree of a spectrum of X-rays depending on an incident angle of the X-rays at each position being highly accurately being reflected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] FIG. **1** is a schematic diagram showing an entire configuration of a phase contrast X-ray imaging apparatus including an X-ray image processing apparatus according to one embodiment.
[0015] FIG. **2** is a schematic view showing a configuration of a grating position adjustment mechanism of the phase contrast X-ray imaging apparatus according to the one embodiment.
[0016] FIG. **3** is a schematic diagram illustrating a configuration of the X-ray image processing apparatus that generates phase contrast X-ray images according to the one embodiment.
[0017] FIG. **4** is a schematic view illustrating uneven noise appearing in a dark field image.
[0018] FIG. **5** is a graph illustrating how a pixel value varies depending on a position along a dashed line in FIG. **4** in the dark field image.
[0019] FIG. **6** is a schematic view illustrating difference between incident angles of X-rays on a grating.
[0020] FIG. **7** is a graph illustrating change of a spectrum of X-rays caused by the difference between incident angles of the X-rays on the grating.
[0021] FIG. **8** is a schematic view illustrating a shape and an arrangement of a correction phantom.
[0022] FIG. **9** is a schematic view illustrating thickness difference of the correction phantom.
[0023] FIGS. **10**(A) and **10**(B) are schematic diagrams illustrating a configuration for generating a correction curve.
[0024] FIG. **11** is a graph illustrating a correction-curve set including a plurality of correction curves.
[0025] FIG. **12** is a schematic diagram illustrating a configuration for generating different types of a plurality of correction-curve sets by using correction phantoms of different materials and different imaging conditions.
[0026] FIGS. **13**(A) and **13**(B) are schematic diagrams illustrating a configuration in which a subject image is corrected by a controller according to the one embodiment, and FIG. **13**(C) is a graph illustrating the correction.
[0027] FIGS. **14**(A), **14**(C) and **14**(E) are schematic diagrams illustrating difference of correction accuracy between a dark field image after correction in a comparative example and a dark field image after correction in the one embodiment, and FIGS. **14**(B), **14**(D) and **14**(F) are graphs illustrating the difference of correction accuracy.
[0028] FIG. **15** is a flowchart illustrating processing of the X-ray image processing apparatus for generating correction curves according to the one embodiment.
[0029] FIG. **16** is a flowchart illustrating processing of the X-ray image processing apparatus for correcting a subject image according to the one embodiment.
[0030] FIG. **17** is a schematic view illustrating a correction phantom according to a first modified embodiment.
[0031] FIG. **18** is a schematic view illustrating a correction phantom according to a second

modified embodiment.

[0032] FIGS. **19**(A) and **19**(B) are schematic views illustrating difference between energy distributions of X-rays depending on orientations of the grating.

MODES FOR CARRYING OUT THE INVENTION

[0033] Embodiments embodying the present invention will be described with reference to the drawings.

[0034] An entire configuration of a phase contrast X-ray imaging apparatus **100** including an X-ray image processing apparatus **6** according to one embodiment of the present invention is now described with reference to FIG. **1**.

[0035] As shown in FIG. **1**, the phase contrast X-ray imaging apparatus **100** is an apparatus for imaging an interior of a subject **90** by using the Talbot effect. The phase contrast X-ray imaging apparatus **100** includes an X-ray source **1**, an X-ray detector **2**, a plurality of gratings, and the X-ray image processing apparatus **6**. In addition, the phase contrast X-ray imaging apparatus **100** includes an input acceptor **10**, a display **11**, and a grating position adjustment mechanism **12**.

[0036] In this specification, an upward/downward direction is defined as a Z direction, and upward and downward directions are defined as Z**1** and Z**2** directions, respectively. Also, a direction between the X-ray source **1** and the X-ray detector **2** is defined as an X direction, and one direction is defined as an X**1** direction while another direction is defined as an X**2** direction in the X direction. In an illustration of FIG. **1**, the X direction is a direction perpendicular to the Z direction. Also, a direction orthogonal to the Z direction and the X direction is defined as a Y direction, and one direction is defined as a Y**1** direction while another direction is defined as a Y**2** direction. The illustration of FIG. **1** is a plan view of the phase contrast X-ray imaging apparatus **100**.

[0037] The X-ray source **1** is configured to irradiate the subject **90** with X-rays. Specifically, the X-ray source **1** is configured to generate X-rays when a high voltage is applied.

[0038] The X-ray detector **2** is configured to detect X-rays with which the subject is irradiated by the X-ray source **1**. The X-ray detector **2** is configured to convert the detected X-rays into electrical signals. The X-ray detector **2** is a flat panel detector (FPD), for example. The X-ray detector **2** includes a plurality of converters (not shown), and a plurality of pixel electrodes (not shown) arranged on the plurality of converters. The plurality of converters and the plurality of pixel electrodes are aligned at a predetermined period (pixel pitch) in the Y and Z directions. The detection signals (image signals) of the X-ray detector **2** are transmitted to the X-ray image processing apparatus **6**.

[0039] The plurality of gratings are arranged between the X-ray source **1** and the X-ray detector **2**. In this embodiment, the plurality of gratings include a first grating **3**, a second grating **4**, and a third grating **5**.

[0040] The first grating **3** is arranged between the X-ray source **1** and the second grating **4**. The first grating **3** is irradiated with X-rays from the X-ray source **1**. The first grating **3** has a plurality of slits **3***a* and X-ray absorption parts **3***b*, which are aligned at a predetermined cycle (pitch) **3***c* in the Z direction. The slits **3***a* and the X-ray absorption parts **3***b* are formed to linearly extend in the Y direction. Also, the slits **3***a* and the X-ray absorption parts **3***b* are formed to extend parallel to each other. The first grating **3** is configured to change X-rays into stripes of X-rays corresponding to positions of the slits **3***a* as a stripe-pattern light source after the X-rays pass through the slits **3***a*.

[0041] The second grating **4** is arranged between the first grating **3** and the third grating **5**. The second grating **4** is irradiated with X-rays from the first grating **3**. The second grating **4** has slits **4***a* and X-ray phase change parts **4***b*, which are arranged at a predetermined cycle (grating pitch) **4***c* in the Y direction. The slits **4***a* and the X-ray phase change parts **4***b* are formed to linearly extend in the Z direction. The second grating **4** is a so-called phase grating. The second grating **4** is provided to form a self-image when irradiated with X-rays from the X-ray source **1** (by using the Talbot effect). Here, the Talbot effect refers to formation of an image of a grating (self-image) having slits at a certain distance (Talbot distance) from the grating when coherent X-rays pass through the

grating.
[0042] The third grating **5** is irradiated with X-rays from the second grating **4**. The third grating **5** has a plurality of X-ray transmission parts **5***a* and X-ray absorption parts **5***b*, which are aligned at a predetermined cycle (grating pitch) **5***c* in the Y direction. The X-ray transmission parts **5***a* and the X-ray absorption parts **5***b* are formed to linearly extend in the Z direction. The third grating **5** is a so-called absorption grating. The third grating **5** is arranged between the second grating **4** and the X-ray detector **2**, and is configured to interfere with the self-image formed by the second grating **4**. To interfere the self-image with the third grating **5**, the third grating **5** is position at the Talbot distance away from the second grating **4**.
[0043] In this embodiment, the first grating **3**, the second grating **4**, and the third grating **5** are orientated to make their grating directions agree with the Z direction. The grating direction is a direction in which a grating pattern forming part extends. The grating pattern forming part in the first grating **3** is a part that is constructed of the slits **3***a* and the X-ray absorption parts **3***b*, and serves as a grating. The grating pattern forming part in the second grating **4** is constructed of the slits **4***a* and the X-ray phase change parts **4***b*. The grating pattern forming part in the third grating **5** is constructed of the X-ray transmission parts **5***a* and the X-ray absorption parts **5***b*.
[0044] The X-ray image processing apparatus **6** includes a processor **7**, such as a CPU (Central Processing Unit), a GPU (Graphics Processing Unit) or an FPGA (Field-Programmable Gate Array) configured for image processing, a memory, such as a ROM (Read Only Memory) and a RAM (Random Access Memory), a storage **8**, and an image acquirer **9**, for example. The X-ray image processing apparatus **6** may include a circuitry instead of the processor **7**.
[0045] The processor **7** includes an image processor **7***a* and a controller **7***b*. The image processor **7***a* is configured to generate a phase contrast X-ray image **30** based on an intensity distribution of X-rays detected by the X-ray detector **2**. The image processor **7***a* is constructed of software as a function block realized by executing the various programs by the processor **7**. The image processor **7***a* may be constructed of hardware as a dedicated processing circuit. In this embodiment, the phase contrast X-ray image **30** includes a first dark field image **35***a* (see FIG. **13**(A)). In this embodiment, the phase contrast X-ray image **30** includes a first absorption image **35***b* (see FIG. **13**(B)). A configuration of the image processor **7***a* that generates the phase contrast X-ray images **30** will be described in detail later.
[0046] The controller **7***b* is configured to control the X-ray source **1**, the grating position adjustment mechanism **12**, and the like. The controller **7***b* is constructed of software as a function block realized by executing the various programs by the processor **7**. The controller **7***b* may be constructed of hardware as a dedicated processing circuit.
[0047] The storage **8** is configured to store a plurality types of correction curve sets **20**, which will be described later. Each correction curve set **20** includes a plurality types of correction curves **20***a*, which will be discussed later. Also, the storage **8** is configured to store the phase contrast X-ray images **30** generated by the image processor **7***a*, and various programs to be executed by the processor **7**. The storage **8** includes a storage device such as an HDD (Hard Disk Drive) or a nonvolatile memory e.g., an SSD (Solid State Drive), for example.
[0048] The image acquirer **9** is configured to acquire the phase contrast X-ray image **30** including the first dark field image **35***a*. The image acquirer **9** is an input/output interface, for example.
[0049] The input acceptor **10** is configured to accept an operating input **21** from a user. The input acceptor **10** includes an input device such as a keyboard, computer mouse, etc. for example.
[0050] The display **11** displays the phase contrast X-ray image **30** generated by the image processor **7***a*. For example, the display **11** includes a display device such as an LCD monitor, an organic EL (Electro Luminescence) monitor, or the like.
[0051] The grating position adjustment mechanism **12** is configured to be able to move the second grating **4** in the X, Y and Z directions, in a rotation direction Rz about an axis extending in the Z direction, a rotation direction Rx about an axis extending in the X direction, and a rotation direction

Ry about an axis extending in the Y direction. A detailed configuration of the grating position adjustment mechanism **12** will be described later.

(Grating Position Adjustment Mechanism)

[0052] As shown in FIG. **2**, the grating position adjustment mechanism **12** includes a Y-directional linear motion mechanism **12***a*, a Z-directional linear motion mechanism **12***b*, and an X-directional linear motion mechanism **12***c*, a linear motion mechanism connection part **12***d*, a stage support driver **12***e*, a stage support **12***f*, a stage driver **12***g*, and a stage **12***h*.

[0053] The Y-directional linear motion mechanism **12***a*, the Z-directional linear motion mechanism **12***b*, and the X-directional linear motion mechanism **12***c* are configured to move in the Y direction, the Z direction and the X direction, respectively. The Y-directional linear motion mechanism **12***a*, the Z-directional linear motion mechanism **12***b*, and the X-directional linear motion mechanism **12***c* include a driver such as a stepping motor, a drive force transmitter for transmitting a drive force from the driver, and a mover to be moved by the drive force transmitted by the drive force transmitter, for example. The grating position adjustment mechanism **12** is configured to move the second grating **4** by operating Y-directional linear motion mechanism **12***a*, the Z-directional linear motion mechanism **12***b*, and the X-directional linear motion mechanism **12***c* in the Y direction, the Z direction and the X direction, respectively.

[0054] The stage support **12***f* supports the stage **12***h* for mounting the second grating **4** from a lower side (Z**2**-direction side) of FIG. **2**. The stage driver **12***g* is configured to be able to move the stage **12***h* back and forth in the Y direction. The stage **12***h* has a bottom surface formed in a convex surface bulging toward the stage support **12***f*, and is configured to rotate about the axis extending in the X direction (to rotate in the Rx direction) when moved in the Y direction. Also, the stage support driver **12***e* is configured be able to move the stage support **12***f* back and forth in the X direction. Also, the stage support **12***f* has a bottom surface formed in a convex surface bulging toward the linear motion mechanism connection part **12***d*, and is configured to rotate about the axis extending in the Y direction (to rotate in the Ry direction) when moved in the X direction. Also, the linear motion mechanism connection part **12***d* is provided to the X-directional linear motion mechanism **12***c* to be able to rotate about the axis extending in the Z direction (to rotate in the Rz direction). According to the aforementioned configuration, the grating position adjustment mechanism **12** can move the second grating **4** in a series of steps (translationally move the second grating) in the X direction, the Y direction, and the Z direction.

(Configuration of Phase Contrast X-Ray Image Generation)

[0055] Processes executed by the image processor **7***a* (see FIG. **1**) to generate the phase contrast X-ray image **30** (see FIG. **1**) is now described with reference to FIG. **3**. The image processor **7***a* generates the phase contrast X-ray image **30** using an intensity signal curve **23** and an intensity signal curve **24** acquired based on an intensity distribution of X-rays detected by the X-ray detector **2** (see FIG. **1**) while translationally moving the second grating **4** by using the grating position adjustment mechanism **12** (see FIG. **1**). The phase contrast X-ray image **30** includes a first absorption image **35***b* (see FIG. **13**(B)), a second absorption image **32** (see FIG. **10**(B)), the phase differential image, a first dark field image **35***a* (see FIG. **13**(A)), and a second dark field image **31** (see FIG. **10**(A)). The intensity signal curve **23** is a curve representing a distribution of intensity of X-rays acquired by capturing an image with the subject **90** (see FIG. **1**) being placed. The intensity signal curve **24** is a curve representing a distribution of intensity of X-rays acquired by capturing an image without the subject **90** being placed. In this embodiment, description of the phase differential image is omitted. The first dark field image **35***a* and the first absorption image **35***b* are a dark field image of the subject **90** captured and an absorption image of the subject **90** captured, respectively. The second dark field image **31** and the second absorption image **32** are a dark field image of a correction phantom **13** (see FIG. **8**) captured and an absorption image of the correction phantom **13** captured, respectively.

[0056] As shown in FIG. **3**, the second absorption image **32** can be generated based on a ratio of average intensity Cs of X-rays when the image is captured with the subject **90** being placed to average intensity Cr of X-rays when the image is captured without the subject **90** being placed. Also, the second dark field image **31** can be generated based on a ratio of Visibility (Vr) of X-rays when the image is captured without the subject **90** being placed to Visibility (Vs) of X-rays when the image is captured with the subject **90** being placed. Vr can be calculated by a ratio of an amplitude Ar of the intensity signal curve **24** to the average intensity Cr. Vs can be calculated by a ratio of an amplitude As of the intensity signal curve **23** to the average intensity Cs.

(Noise Appearing in Dark Field Image)

[0057] The following description describes uneven noise that appears in the second dark field image **31** (see FIG. **4**) with reference to FIGS. **4** to **7**. Here, difference of pixel values is represented by different hatching patterns in the second dark field image **31** shown in FIG. **4**. Specifically, the thinner the hatching pattern (the wider the distance between hatching lines), the larger the pixel value. Here, illustrations of FIGS. **4** to **7** represent the uneven noise that appears in the second dark field image **31** generated by the phase contrast X-ray imaging apparatus **100** according to this embodiment, and is to be reduced by correction processing, which will be described later.

[0058] The illustration of FIG. **4** is a second dark field image **31** when an image of a flat plate-like aluminum material as the subject **90** is captured. The plate-like aluminum material does not normally induce X-ray scattering so that the plate-like aluminum material will not be imaged in the second dark field image **31**. However, when an image of the plate-like aluminum material is captured, the subject **90** that includes pixel values different from a background **91** is imaged in the second dark field image **31** as shown in FIG. **4**. This can be conceived due to change (hardening) of a spectrum of X-rays when the X-rays pass through the subject **90**. Here, hardening of a spectrum of X-rays means that a central energy of the spectrum of the X-rays shifts to a higher energy side.

[0059] Also, as shown in FIG. **4**, an area **90***a* in a central part in the Y direction of the subject **90** includes a pixel value higher than an area **90***b* and an area **90***c* on both sides in the second dark field image **31**.

[0060] FIG. **5** is a graph **50** of pixel values plotted along a line **40** extending in the Y direction in the second dark field image **31** shown in FIG. **4**. A vertical axis indicates the pixel value, and a horizontal axis indicates positions (pixels) in the graph **50**.

[0061] As shown by a curve **50***a* in the graph **50**, it is confirmed that the pixel value of area **90***a* (see FIG. **4**) in the central part of the second dark field image **31** (see FIG. **4**) is higher than the pixel values of the area **90***b* (see FIG. **4**) and the area **90***c* (see FIG. **4**) on both sides.

[0062] Here, X-rays for irradiation from the X-ray source **1** (see FIG. **6**) are so-called cone-beam X-rays. Accordingly, the X-rays are incident on the plurality of gratings in an inclined direction. Because a hardening degree of a spectrum of X-rays changes due to difference between incident angles of the X-rays on the plurality of gratings, the pixel values of the area **90***a* in the central part of the second dark field image **31** becomes greater than the pixel values of the area **90***b* and the area **90***c* on the both sides.

[0063] An arrow **41** in FIG. **6** indicates X-rays that travel along an optical axis of the X-rays and with which the second grating **4** is irradiated. A dashed arrow **42** indicates X-ray that is incident from on the grating in an inclined direction. Here, in an illustration of FIG. **6**, the second grating **4** is only shown in the plurality of gratings for ease of illustration. A spectrum of X-rays when the X-rays indicated by the arrow **41** pass through the second grating **4** and are detected by the X-ray detector **2** becomes different from a spectrum of X-rays when the X-rays indicated by the arrow **42** pass through the second grating **4** and are detected by the X-ray detector **2**, because path lengths of the X-rays passing through the second grating **4** are different due to difference between incident angles of the X-rays on the second grating **4**.

[0064] A graph **51** shown in FIG. **7** represents a spectrum of X-rays that pass through the second grating **4** (see FIG. **6**). A vertical axis indicates X-ray intensity, and a horizontal axis indicates X-ray energy in the graph **51**.

[0065] A curve **51***a* shown in the graph **51** represents a spectrum of the X-rays indicated by the arrow **41** in FIG. **6**. A dashed curve **51***b* in the graph **51** represents a spectrum of the X-rays indicated by the dashed arrow **42** in FIG. **6**.
[0066] In a case in which X-rays are incident on the second grating **4** in an inclined direction, a path length of the X-rays indicated by the arrow **42** for passing through the second grating **4** is greater than a path length of the X-rays indicated by the arrow **41**. Because the path lengths are different, the spectrum of the X-rays represented by the curve **51***b* after passage through the second grating **4** becomes harder than the spectrum of the X-rays represented by the curve **51***a*. Because a hardening degree of a spectrum of X-rays changes in accordance with an incident angle of the X-rays on the second grating **4**, uneven noise appears in a direction (Y direction) perpendicular to the grating direction (Z direction) in the second dark field image **31**. If uneven noise appears in the second dark field image **31**, a profile of pixel values has a convex shape having a center part bulging upward as shown by the curve **50***a* in the graph **50** shown in FIG. **5**.
[0067] In a case in which an image of the subject **90** is captured by using an X-ray image processing apparatus having a plurality of gratings as in the phase contrast X-ray imaging apparatus **100** according to this embodiment, a change of a spectrum of X-rays due to the subject **90** and a change of the spectrum of the X-rays due to difference between incident angles of the X-rays on the second grating **4** occur. For this reason, as shown in FIG. **4**, uneven noise appears in the second dark field image **31**, and image quality of the second dark field image **31** deteriorates. The uneven noise appearing in the second dark field image **31** is caused by a combination of the change of the spectrum of the X-rays due to the subject **90** and the change of the spectrum of the X-rays due to the difference between incident angles of the X-rays on the second grating **4**.
[0068] To address this, in this embodiment, the controller **7***b* is configured to correct the first dark field image **35***a* based on a plurality of correction curves **20***a*, which are generated at positions in the direction (Y direction) perpendicular to the grating direction (Z direction), and represents assignment relations between values that relate to X-ray absorption of the subject **90** and values that relate to X-ray scattering (scattering degree) of the subject at each of the positions. Each of the plurality of correction curves **20***a* are generated based on a correction image **33**. The correction image **33** is an image of the correction phantom **13** captured.
(Correction Phantom)
[0069] The correction phantom **13** used to generate the plurality of correction curves **20***a* is first described with reference to FIGS. **8** and **9**. Here, the gratings are arranged with their grating directions are the same direction as each other, in an illustration of FIG. **8**, only the second grating **4** is shown for ease of illustration.
[0070] As shown in FIG. **8**, the correction phantom **13** has a structure whose X-ray absorption degree changes in a direction of the grating extension (Z direction). The correction phantom **13** is formed to change its thickness in an X-ray irradiation direction (X direction) in the Z direction. Specifically, the correction phantom **13** has a stepped shape in which its thickness in the X-ray irradiation direction changes in steps in the Z direction. The correction phantom **13** is integrally formed of a homogeneous material.
[0071] The correction phantom **13** has a plurality of steps. Specifically, the correction phantom **13** has a first step **13***a* to a fourth step **13***d*.
[0072] As shown in FIG. **9**, in thicknesses of the steps, a thickness **60***a* of the first step **13***a* is the largest, and a thickness **60***d* of the fourth step **13***d* is a the smallest.
[0073] A thickness per step of the correction phantom **13** is constant. In other words, with reference to the thickness **60***d* of the fourth step **13***d*, the thickness **60***c* of the third step **13***c* is twice the thickness **60***d*. Similarly, the thickness **60***b* of the second step **13***b* is triple the thickness **60***d*. Also, the thickness **60***a* of the first step **13***a* is four times the thickness **60***d*. Note that a thickness per step of the correction phantom **13** may not be constant.
(Correction Image)

[0074] Correction images **33** as shown in FIGS. **10**(A) and **10**(B) are obtained by capturing an image of the correction phantom **13**.

[0075] FIG. **10**(A) is a schematic diagram showing the second dark field image **31**, which is a dark field image of the correction phantom **13**, in the correction images **33**. Also, FIG. **10**(B) is a schematic diagram showing the second absorption image **32**, which is an absorption image of the correction phantom **13**, in the correction images **33**. Here, difference of pixel values is represented by different hatching patterns in FIGS. **10**(A) and **10**(B). Specifically, the denser the hatching pattern (the narrower the distance between hatching lines), the larger the pixel value in FIGS. **10**(A) and **10**(B).

[0076] In the second dark field image **31** shown in FIG. **10**(A), areas having different pixel values imaged in accordance with the thicknesses of the correction phantom **13**. Specifically, a first area **70***a* is the area where the first step **13***a* of the correction phantom **13** is imaged. Also, a second area **70***b* is the area where the second step **13***b* of the correction phantom **13** is imaged. Also, a third area **70***c* is the area where the third step **13***c* of the correction phantom **13** is imaged. Also, a fourth area **70***d* is the area where the fourth step **13***d* of the correction phantom **13** is imaged. Also, a fifth area **70***e* is the area where no correction phantom **13** is imaged (background).

[0077] In the second absorption image **32** shown in FIG. **10**(B), areas having different pixel values depending on the thicknesses of the correction phantom **13** are imaged. Specifically, a first area **71***a* is the area where the first step **13***a* of the correction phantom **13** is imaged. Also, a second area **71***b* is the area where the second step **13***b* of the correction phantom **13** is imaged. Also, a third area **71***c* is the area where the third step **13***c* of the correction phantom **13** is imaged. Also, a fourth area **71***d* is the area where the fourth step **13***d* of the correction phantom **13** is imaged. Also, a fifth area **71***e* is the area where no correction phantom **13** is imaged (background).

[0078] As shown in FIGS. **10**(A) and **10**(B), uneven noise due to hardening of a spectrum of X-rays after passage through the subject **90**, and hardening of a spectrum of the X-rays due to difference between incident angles of the X-rays on the plurality of gratings appears only in the second dark field image **31**.

(Correction Curve)

[0079] A configuration of the controller **7***b* that generates the plurality of correction curves **20***a* is now described with reference to FIGS. **10** to **13**.

[0080] In this embodiment, the plurality of correction curves **20***a* are generated based on pixel values in the second absorption image **32** and pixel values in the second dark field image **31** of the areas that have different thicknesses from each other of the correction phantom **13** captured in the correction image **33** with the correction phantom **13** being orientated to change its thickness in the grating direction (Z direction).

[0081] In other words, the controller **7***b* acquires an assignment relation between the pixel values of the second dark field image **31** (values that relate to X-ray scattering and are specified based on the second dark field image **31**), and pixel values of the second absorption image **32** (values that relate to X-ray absorption and are specified based on the second absorption image **32**), and generates the plurality of correction curves **20***a*. Accordingly, because the correction phantom **13** is formed of a homogeneous material, in a case in which a thickness of the correction phantom **13** is fixed, change of the pixel values of the second dark field image **31** with respect to the pixel values of the second absorption image **32** becomes constant. That is, a factor of the change of the pixel values of the second dark field image **31** with respect to the pixel values of the second absorption image **32** is limited to the difference between the thicknesses of the correction phantom **13**. As a result, the correction curves **20***a* corresponding to the difference between the thicknesses of the subject **90** can be easily generated by generating the correction curves **20***a* by using the pixel values in the second absorption image **32** and the pixel values in the second dark field image **31** of the areas of the correction phantom **13** that have different thicknesses from each other depending on positions in the direction perpendicular to the grating direction.

[0082] FIG. **11** is a graph **52** showing the plurality of correction curves **20***a*. The graph **52** includes graphs aligned at their corresponding Y-directional positions in the correction image **33**, and each of the graphs has a vertical axis indicating absorption values (exemplary values that relate to X-ray absorption) and a horizontal axis indicating dark field values (exemplary values that relate to X-ray scattering). Here, each of the dark field values and each of absorption values are acquired by the following Equations (1) and (2).

[Formula 1]

[00001] $= \ln D \quad (1) \qquad = \ln T \quad (2)$

[0083] where α is the dark field value, D is a pixel value of the second dark field image **31**, γ is the absorption value, and T is a pixel value of the second absorption image **32**. Here, the dark field and absorption values are expressed in natural logarithms.

[0084] The configuration of the controller **7***b* that generates the plurality of correction curves **20***a* is described by describing a configuration of the controller **7***b* that generates a curve **52***a* shown in the graph **52** as a representative.

[0085] In this embodiment, the controller **7***b* determines a position of a first plot **14***a* in the graph **52** based on the dark field value acquired in accordance with the pixel value of a first pixel **31***a* (see FIG. **10**(A)) in the second dark field image **31** and the absorption value acquired in accordance with the pixel value of a first pixel **32***a* (see FIG. **10**(B)) in the second absorption image **32**. Also, the controller **7***b* determines a position of a second plot **14***b* in the graph **52** based on the dark field value acquired in accordance with the pixel value of a second pixel **31***b* (see FIG. **10**(A)) in the second dark field image **31** and the absorption value acquired in accordance with the pixel value of a second pixel **32***b* (see FIG. **10**(B)) in the second absorption image **32**.

[0086] Also, the controller **7***b* determines a position of a third plot **14***c* in the graph **52** based on the dark field value acquired in accordance with the pixel value of a third pixel **31***c* (see FIG. **10**(A)) in the second dark field image **31** and the absorption value acquired in accordance with the pixel value of a third pixel **32***c* (see FIG. **10**(B)) in the second absorption image **32**. Also, the controller **7***b* determines a position of a fourth plot **14***d* in the graph **52** based on the dark field value acquired in accordance with the pixel value of a fourth pixel **31***d* (see FIG. **10**(A)) in the second dark field image **31** and the absorption value acquired in accordance with the pixel value of a fourth pixel **32***d* (see FIG. **10**(B)) in the second absorption image **32**.

[0087] Here, the correction image **33** includes random noise in some cases. In such a case, if a correction curve **20***a* is generated based on a pixel value of one pixel in the area of the correction phantom **13** that have the same thickness, accuracy of the correction curve **20***a* (approximation accuracy) may be reduced due to the random noise. To address this, in this embodiment, each of the plurality of correction curves **20***a* is generated based on pixel values of the second absorption image **32** and pixel values of the second dark field image **31** at different positions in the area that has the same thickness of the grating direction in the correction phantom **13** captured in the correction image **33**. In other words, each of the plurality of correction curves **20***a* is generated based on representative values of pixel values of the second dark field image **31** and representative values of pixel values of the second absorption image **32**. Each representative value is an average value of the pixel values in the same area that extends in the Y direction and has the same thickness, for example. That is, in a case in which the average value of the first pixel **31***a* is acquired, an average value of pixel values at a Y-directional position Ya in the first area **70***a* is acquired. As a result, the correction curve **20***a* can be generated by using representative pixel values of the second absorption image **32** each of which corresponds to a plurality of positions in each area that has the same thickness in the grating direction, and representative pixel values of the second dark field image **31** each of which corresponds to a plurality of positions in each area that has the same thickness in the grating direction, for example. Consequently, it is possible to suppress reduction of accuracy of the correction curve **20***a* that is caused by random noise appearing in the correction image **33** as compared with a configuration in which the correction curve **20***a* is generated based on pixel values of the second absorption image **32** and pixel values of the second dark field image **31** each of which corresponds to one position in each area in the correction phantom **13** that has the same thickness.

[0088] The controller **7***b* generates the correction curve **20***a* that is indicated by the curve **52***a* by approximation of the first plot **14***a* to the fourth plot **14***d* to a curve.

[0089] The controller **7***b* generates the correction curve **20***a* that is indicated by a curve **52***b*, and the correction curve **20***a* that is indicated by a curve **52***c* at Y-directional positions in the correction image **33** by using similar processing by changing one Y-directional position of the correction image **33** to another. That is, the controller **7***b* generates the correction curve **20***a* at a Y-directional position Ya, the correction curve **20***a* at Y-directional position Yb, and the correction curve **20***a* at Y-directional position Yc.

[0090] Here, change of pixel values of the second dark field image **31** in the Y direction is a gradual change, it is not necessary to generate the correction curve **20***a* at every pixel in the Y direction. For this reason, the controller **7***b* generates a predetermined number of correction curves **20***a* depending on Y-directional size of the correction image **33**. In illustrations of FIGS. **10** and **11**, the controller **7***b* generates three correction curves **20***a*. Note that the number of correction curves **20***a* to be generated may be any number other than three. For example, the number of correction curves **20***a* may be determined based on a profile of pixel values of the second dark field image **31** in the Y direction (based on the graph **50** shown in FIG. **5**), or a correction curve **20***a* may be generated by each predetermined distance in the Y direction.

[0091] Here, transmittance of X-rays varies in accordance with X-ray absorptance and thickness of the subject **90**. Accordingly, even in a case in which a thickness of the subject **90** and a thickness of the correction phantom **13** are equal to each other, if their X-ray absorptances are different, the difference between X-ray absorptances may reduce accuracy of the correction of the first dark field image **35***a* using the correction curves **20***a*. For this reason, it is preferable to change the correction phantom **13** to be used to generate the correction curves **20***a* depending on a material of the subject **90**. Specifically, the correction phantom **13** is preferably changed depending on an elemental composition of the subject **90**. For example, in a case in which the subject **90** is CFRP (Carbon Fiber Reinforced Plastics), it is preferable to use the correction phantom **13** that is formed by an acrylic material because the acrylic material has an X-ray absorptance close to CFRP and a homogeneous phantom of the acrylic material that has few defects and a small density variation can be easily obtained. Also, in a case in which the subject **90** is GFRP (Glass Fiber Reinforced Plastics), it is preferable to use the correction phantom **13** that is formed by an aluminum material because the aluminum material has an X-ray absorptance close to GFRP and a homogeneous phantom of the aluminum material that has few defects and a small density variation can be easily obtained.

[0092] For this reason, in this embodiment, depending on materials of the different subjects **90**, images of different correction phantoms **13** that have different X-ray absorptances are captured so that a plurality types of correction curve sets **20** are generated based on a plurality of correction image sets **34** (see FIG. **12**) corresponding to different types of correction phantoms **13**, and are included as the plurality of correction curves **20***a*.

[0093] Also, in a case in which setting of imaging conditions **22** for capturing the phase contrast X-ray image **30** is changed, change degrees of spectra of X-rays become different between before and after the changing of the imaging conditions **22**. To address this, in this embodiment, the plurality of correction curves **20***a* include a plurality of correction curve sets **20**, which are generated based on the correction images **33** captured under imaging conditions **22** different from each other, for the imaging conditions **22**. Here, the imaging conditions **22** include a value of tube voltage applied to the X-ray source **1**, and a rotation angle of the plurality of gratings.

[0094] A configuration of the controller **7***b* that generates the plurality types of correction curve sets **20** is now described with reference to FIG. **12**.

[0095] In an illustration of FIG. **12**, a first correction phantom **13***e* and a second correction phantom **13***f* are used to generate the plurality types of correction curve sets **20**. The first correction phantom **13***e* is a phantom formed of an acrylic material, for example. Also, the second correction phantom **13***f* is a phantom formed of an aluminum material, for example.

[0096] Also, first and second imaging conditions are imaging conditions **22** that tube voltages applied to the X-ray source **1** are different from each other.

[0097] In the illustration of FIG. **12**, the plurality of correction image sets **34** include the correction image **33** of the first correction phantom **13***e* that is captured under the first and second imaging conditions, and the correction image **33** of the second correction phantom **13***f* that is captured under the first and second imaging conditions so that the plurality of correction image sets include a total of the four correction images **33**. Each of the four correction images **33** includes the dark field image **33***a* and the absorption image **33***b*.

[0098] The controller **7***b* generates the plurality of correction curve sets **20** using their respective four types of correction images **33**. Subsequently, the controller **7***b* stores the plurality of correction curve sets **20** into the storage **8**.

[0099] In this embodiment, the storage **8** stores the plurality of correction curves **20***a* associated with the imaging conditions **22**. Also, the storage **8** stores the plurality of correction curves **20***a* associated with materials of the subjects **90**. That is, in this embodiment, the storage **8** stores a plurality types of correction curve sets **20** associated with the imaging conditions **22** and X-ray absorptances. Specifically, the storage **8** stores the correction curve set **20** that is generated based on the correction image set **34** of the first correction phantom **13***e* that is captured under the first imaging condition, the correction curve set **20** that is generated based on the correction image set **34** of the second correction phantom **13***f* that is captured under the first imaging condition, the correction curve set **20** that is generated based on the correction image set **34** of the first correction phantom **13***e* that is captured under the second imaging condition, and the correction curve set **20** that is generated based on the correction image set **34** of the second correction phantom **13***f* that is captured under the second imaging condition.

[0100] Here, the correction curve **20***a* is a curve that assigns dark field values with absorption values. Accordingly, based on an absorption value acquired based on a value of a pixel of the second absorption image **32** and the correction curve **20***a*, a dark field value of the pixel that is assigned to the absorption value can be acquired.

(Correction of Dark Field Image)

[0101] A configuration of the controller **7***b* that corrects a subject image **35** is now described with reference to FIGS. **13**(A) to **13**(C). Here, the subject image **35** includes the first dark field image **35***a* shown in FIG. **13**(A) and the first absorption image **35***b* shown in FIG. **13**(B). The controller **7***b* corrects the first dark field image **35***a* in the subject image **35**. A graph **53** shown in FIG. **13**(C) shows the correction curve **20***a* that corresponds to a first area **72***a* (see FIG. **13**(A)) in the plurality of correction curves **20***a*. A vertical axis indicates an absorption value, and a horizontal axis indicates a dark field value in the graph **53**. In each of illustrations of FIGS. **13**(A) and **13**(B), an image including areas whose pixel values are different in the Z direction is shown for ease of understanding effects of the correction.

[0102] The controller **7***b* is configured to correct the first dark field image **35***a* by using the correction curves **20***a* corresponding to positions.

[0103] The controller **7***b* is configured to correct the first dark field image **35***a* based on the first absorption image **35***b* and the plurality of correction curves **20***a*. Because uneven noise causes a gradual change of pixel values in the Y direction, the controller **7***b* corrects the subject image **35** in the Z direction, for positions at a predetermined number of pixels from each other in the Y direction, by using corresponding one the correction curves **20***a*. In the illustration of FIG. **13**(A), the controller **7***b* corrects the subject image **35** in the Z direction by using the correction curves **20***a*

corresponding the first area **72**a, the second area **72**b, and the third area **72**c.
[0104] As a representative of the configuration of the controller **7**b that corrects the first dark field image **35**a, correction of the subject image **35** (the first dark field image **35**a) at the first area **72**a is described. A first absorption value **80**a of a first pixel **37**a is acquired by using a pixel value of the first pixel **37**a in a first area **73**a of the first absorption image **35**b corresponding to the first area **72**a of the first dark field image **35**a and the above Equation (2). Subsequently, the controller **7**b acquires a first dark field value **81**a based on the first absorption value **80**a and the correction curve **20**a.
[0105] Similarly, the controller **7**b acquires a second absorption value **80**b, a third absorption value **80**c, and a fourth absorption value **80**d based on a pixel value of a second pixel **37**b, a pixel value of a third pixel **37**c and a pixel value of a fourth pixel **37**d in the first absorption image **35**b, and the above Equation (2). Subsequently, the controller **7**b acquires a second dark field value **81**b, a third dark field value **81**c, and a fourth dark field value **81**d based on the second absorption value **80**b, the third absorption value **80**c and the fourth absorption value **80**d, and the correction curve **20**a. That is, the controller **7**b is configured to access, for each of the positions in the first absorption image **35**b, corresponding one of the correction curves **20**a, and to specify values that relate to the X-ray scattering and are assigned to values that relate to the X-ray absorption in the first absorption image **35**b.
[0106] The controller **7**b acquires pseudo-signal components of pixels from dark field values of the first pixel **36**a to the fourth pixel **36**d of the first dark field image **35**a by using the first dark field value **81**a to the fourth dark field value **81**d acquired, and the following Equation (3). Here, the pseudo-signal components are signal components of the first dark field image **35**a that includes uneven noise.

[00002] [Formula2]

$$D_f = \frac{1}{e} \quad (3)$$

where D.sub.f is a pixel value of each pseudo-signal component, a is the dark field value, and e is the base of the natural logarithm.
[0107] Subsequently, the controller **7**b uses the pixel values of the pseudo-signal components of the pixels and the pixel value of the pixels to acquire pixel values of the pixels in a dark field image after correction **35**d (see FIG. **14**(E)) as shown in the following Equation (4).

[00003] [Formula3]

$$D_{corr} = D / D_f \quad (4)$$

[0108] where D.sub.corr is each pixel value of the dark field image after correction **35**d.
[0109] The controller **7**b corrects the first dark field image **35**a, that is, corrects the second area **72**b and the third area **72**c of the first dark field image **35**a by using the correction curves **20**a corresponding to these areas, and the pixel values of the second area **73**b and the third area **73**c of the first absorption image **35**b. In other words, the controller **7**b is configured to correct the first dark field image **35**a by using the values specified that relate to X-ray scattering.
[0110] Here, in this embodiment, the controller **7**b is configured to select a plurality of correction curves **20**a (see FIG. **1**) in corresponding one set of the plurality types of correction curve sets **20** (see FIG. **1**) depending on a material of the subject **90**. Also, the controller **7**b is configured to correct the first dark field image **35**a (see FIG. **14**(A)) by using the plurality of correction curves **20**a selected. Specifically, the controller **7**b is configured to select the correction curves **20**a to be used to correct the first dark field image **35**a based on the operating input **21** accepted by the input acceptor **10**. For example, the controller **7**b displays a combo box for selecting a type of the subject **90** on the display **11**. The controller **7**b acquires the correction curves **20**a that are used to correct the first dark field image **35**a in accordance with the material of the subject **90** input by an operator by selecting the material in the combo box. Accordingly, the first dark field image **35**a is corrected by using the correction curves **20**a corresponding to the operating input **21** of the operator. As a

result, because the first dark field image **35***a* can be corrected by the correction curves **20***a* that are selected by the operator, it is possible to improve user convenience (usability).

[0111] The controller **7***b* is configured to select a plurality of correction curves **20***a* in accordance with the imaging condition **22**. Also, the controller **7***b* is configured to correct the first dark field image **35***a* (see FIG. **14**(A)) by using the plurality of correction curves **20***a* selected. Specifically, the controller **7***b* acquires a value of tube voltage applied to the X-ray source **1** as the imaging condition **22**. The controller **7***b* acquires a plurality of correction curves **20***a* that are used to correct the first dark field image **35***a* from the plurality types of correction curve sets **20** in accordance with the value of tube voltage acquired.

[0112] In other words, the controller **7***b* acquires the plurality of correction curves **20***a* that are used to correct the first dark field image **35***a* (see FIG. **14**(A)) from the plurality of correction curve sets **20** in accordance with the material of the subject **90** and the imaging condition **22**.

(Effect of Correction of Dark Field Image)

[0113] The following description describes a dark field image after correction **35***c*, which is generated by correcting a subject image before correction (first dark field image **35***a*), in a comparative example, and a dark field image after correction **35***d*, which is generated by correcting the subject image before correction by using the correction curve set **20** corresponding to positions in the Y direction, in this embodiment with reference to FIGS. **14**(A) to **14**(F). The dark field image after correction **35***c* in the comparative example is corrected by using one type of correction curve set **20** independent of positions in the Y direction.

[0114] The first dark field image **35***a* shown in FIG. **14**(A) is a dark field image before correction. A graph **54** of FIG. **14**(B) shows changes of pixel values along a first line **41***a* to a fifth line **41***e* in a first area **74***a* to a fifth area **74***e* of the first dark field image before correction **35***a*, respectively.

[0115] An illustration of FIG. **14**(C) shows the dark field image after correction **35***c*, which is generated by correcting the first dark field image before correction **35***a* by using one type of correction curve set **20** independent of positions in the Y direction, in a comparative example. Also, a graph **55** of FIG. **14**(D) shows changes of pixel values along a first line **42***a* to a fifth line **42***e* in the first area **74***a* to the fifth area **74***e* of the first dark field image after correction **35***c* in the comparative example, respectively.

[0116] An illustration of FIG. **14**(E) shows the dark field image after correction **35***d*, which is generated by correcting the first dark field image before correction **35***a* by using by using the correction curve set **20** corresponding to positions in the Y direction, in this embodiment. Also, a graph **56** of FIG. **14**(F) shows changes of pixel values along a first line **43***a* to a fifth line **43***e* in the first area **74***a* to the fifth area **74***e* of the first dark field image after correction **35***d* in this embodiment, respectively.

[0117] A vertical axis indicates the pixel value, and a horizontal axis indicates the positions (pixels) in the graphs **54** to **56**. A first curve **54***a*, a second curve **54***b*, a third curve **54***c*, a fourth curve **54***d*, and a fifth curve **54***e* shown in the graph **54** show changes of pixel values along the first line **41***a*, the second line **41***b*, the third line **41***c*, the fourth line **41***d*, and the fifth line **41***e*, respectively, in the areas of the dark field image before correction **35***d*.

[0118] Also, a first curve **55***a*, a second curve **55***b*, a third curve **5***c*, a fourth curve **55***d*, and a fifth curve **55***e* shown in the graph **55** show changes of pixel values along the first line **42***a*, the second line **42***b*, the third line **42***c*, the fourth line **42***d*, and the fifth line **42***e*, respectively, in the areas of the dark field image after correction **35***c* in the comparative example.

[0119] Also, a first curve **56***a*, a second curve **56***b*, a third curve **56***c*, a fourth curve **56***d*, and a fifth curve **56***e* shown in the graph **56** show changes of pixel values along the first line **43***a*, the second line **43***b*, the third line **43***c*, the fourth line **43***d*, and the fifth line **43***e*, respectively, in the areas of the dark field image after correction **35***d* in this embodiment.

[0120] In the first dark field image before correction **35***a*, as shown in the graph **54**, the pixel values in the first area **74***a* to the fifth area **74***e* are different from each other. Also, in the second area **74***b* to the fifth area **74***e*, the pixel values in the central part of the image in the Y direction are high, and the pixel values on both sides are low as shown in the graph **54**.

[0121] In the dark field image after correction **35***c* in the comparative example, as shown in the graph **55**, difference between the pixel values in the first area **74***a* to the fifth area **74***e* becomes smaller than difference between the pixel values in the first dark field image before correction **35***a*. In addition, uneven noise is eliminated in the central part of the image. However, because the dark field image after correction **35***c* in the comparative example is corrected by using one type of correction curve set **20** independent of positions in the Y direction, the pixel values in the central part of the image are still high, and the pixel values on both sides are still low the second area **74***b* to the fifth area **74***e* as shown in the graph **55**.

[0122] In the dark field image after correction **35***d* in this embodiment, as shown in the graph **56**, difference between the pixel values in the first area **74***a* to the fifth area **74***e* substantially disappears. In addition, in the dark field image after correction **35***d* in this embodiment, difference between the pixel values in the central parts of the image and the pixel values in the both sides of the image substantially disappears in the second area **74***b* to the fifth area **74***e*. That is, in the dark field image after correction **35***d* in this embodiment, uneven noise due to interaction between both a change of a spectrum of X-rays caused by the subject **90** (see FIG. **1**) and a change of the spectrum of X-rays caused by X-rays incident on the grating in an inclined direction is reduced.

(Generation Processing of Correction Curves and Correction Processing of Dark Field Image)

[0123] The following description describes a configuration of an X-ray image processing method by using the X-ray image processing apparatus **6** according to this embodiment with reference to FIGS. **15** and **16**. In this embodiment, the X-ray image processing method includes processing for generating the correction curves **20***a* and processing for correcting the first dark field image **35***a*, which are main divided parts of the method.

[0124] The processing for generating the correction curves **20***a* by using the X-ray image processing apparatus **6** is first described with reference to FIG. **15**.

[0125] In step **101**, the controller **7***b* captures the correction image **33** including the second dark field image **31** (see FIG. **10**(A)) and the second absorption image **32** (see FIG. **10**(B)) of the correction phantom **13** captured by controlling the X-ray source **1**.

[0126] The controller **7***b* captures the correction image **33** with the correction phantom **13** being orientated to make a direction in which a thickness of the correction phantom **13** changes agree with the grating direction (Z direction).

[0127] In step **102**, the controller **7***b* generates the correction curves **20***a* based on the correction image **33**.

[0128] Here, in step **102**, the controller **7***b* generates the correction curves **20***a* based on pixel values in the second absorption image **32** and pixel values in the second dark field image **31** of the areas that have different thicknesses from each other of the correction phantom **13** captured in the correction image **33**. Specifically, the controller **7***b* generates plurality of correction curves **20***a* based on pixel values in the second absorption image **32** and pixel values in the second dark field image **31** at different positions in the direction (Y direction) perpendicular to the grating direction (Z direction). More specifically, the controller **7***b* generates each of the plurality of correction curves **20***a* based on pixel values of the second absorption image **32** and pixel values of the second dark field image **31** at different positions in the area that has the same thickness of the grating direction in the correction phantom **13** captured in the correction image **33**. Also, the controller **7***b* stores plurality of correction curves **20***a* generated into the storage **8**.

[0129] In step **103**, the controller **7***b* determines whether the plurality of correction curves **20***a* are generated under all of the predetermined imaging conditions **22**. If the plurality of correction curves **20***a* are not generated under all of the predetermined imaging conditions **22**, the procedure goes to step **104**. If the plurality of correction curves **20***a* are generated under all of the predetermined imaging conditions **22**, the procedure goes to step **105**.

[0130] In step **104**, controller **7***b* changes one of the imaging conditions **22** to another. Subsequently, the procedure goes to step **101**.
[0131] If the procedure goes from step **103** to step **105**, in step **105**, the controller **7***b* determines whether the plurality of correction curves **20***a* are generated in all of the plurality of correction phantoms **13** (see FIG. **8**). If the plurality of correction curves **20***a* are not generated in all of the plurality of correction phantoms **13**, the procedure goes to step **106**. If the plurality of correction curves **20***a* are generated in all of the plurality of correction phantoms **13**, the procedure ends.
[0132] In step **106**, controller **7***b* changes one of the correction phantoms **13** to another. For example, the controller **7***b* urges operators to change the correction phantom **13** by displaying a message to change the correction phantom **13** on the display **11**. If the correction phantom **13** is changed, the procedure goes to step **101**.
[0133] As described above, each of the plurality of correction curve sets **20** is generated under corresponding one of the imaging conditions **22** in steps for capturing an image of the correction phantom **13** corresponding one of the imaging conditions **22** different from each other and generating the plurality of correction curves **20***a* by repeating steps **101** to **104**.
[0134] Also, each of the plurality of correction curve sets **20** is generated in corresponding one of the plurality of types of correction phantoms **13** in steps for capturing an image of the corresponding one of the plurality of types of correction phantoms **13**, which have X-ray absorptances different from each other, depending on materials of the different subjects **90** and generating the plurality of correction curves **20***a* by repeating steps **101** to **106**.
[0135] The processing for correcting the subject image **35** by using the X-ray image processing apparatus **6** is now described with reference to FIG. **16**.
[0136] In step **200**, the image acquirer **9** acquires the subject image **35** of the subject **90** captured including the first dark field image **35***a* (see FIG. **13**(A)) and the first absorption image **35***b* (see FIG. **13**(B)).
[0137] In step **201**, the controller **7***b* acquires the imaging condition **22**. In this embodiment, the controller **7***b* acquires a value of tube voltage applied to the X-ray source **1** as the imaging condition **22**.
[0138] In step **202**, the controller **7***b* acquires the material of the subject **90**. Specifically, the controller **7***b* accepts an input of the material of the subject **90**. The controller **7***b* accepts the input of the material of the subject **90** by accepting the operating input **21** through the input acceptor **10**.
[0139] In step **203**, the controller **7***b* acquires the correction curve set **20** that is used to correct the first dark field image **35***a* from the plurality types of correction curve sets **20** based on the imaging condition **22** and the material of the subject **90**.
[0140] In step **204**, the controller **7***b* corrects the subject image **35** (first dark field image **35***a*) based on the plurality of correction curves **20***a*. Specifically, the controller **7***b* corrects the first dark field image **35***a* based on the first absorption image **35***b* (see FIG. **13**(B)) and the plurality of correction curves **20***a*. More specifically, the controller **7***b* corrects the subject image **35** (first dark field image **35***a*) based on the plurality of correction curves **20***a* corresponding to the material of the subject **90** accepted as the input in the plurality of correction curve sets **20**. The controller **7***b* corrects the subject image **35** (first dark field image **35***a*) based on the plurality of correction curves **20***a* corresponding to the acquired imaging condition **22** in the plurality of correction curve sets **20**. In other words, the controller **7***b* determines the correction curve set **20** to be used for the correction of the subject image **35** from the plurality of correction curve sets **20** based on the material of the subject **90** and the imaging condition **22**, and corrects the subject image **35**. After that, the procedure ends.
Advantages of the Embodiment
[0141] In this embodiment, the following advantages are obtained.
[0142] In the aforementioned embodiment, the phase contrast X-ray imaging apparatus **100** includes an X-ray source **1**; an X-ray detector **2** for detecting the X-rays for irradiation from the X-

ray source **1**; a plurality of gratings arranged between the X-ray source and the X-ray detector **2**; an image processor **7***a* for generating a phase contrast X-ray image **30** including a first dark field image **35***a*, which is a dark field image of a subject **90**, based on an intensity distribution of the X-rays detected by the X-ray detector **2**; a storage **8** for storing a plurality of correction curves **20***a* generated at positions in a direction perpendicular to a grating direction in which the plurality of gratings extend; and a controller **7***b* for correcting the first dark field image **35***a* by using the correction curves **20***a* corresponding to the positions, wherein each of the correction curves **20***a* represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject **90** at corresponding one of the positions.

[0143] Accordingly, the storage **8** stores a plurality of correction curves **20***a* corresponding to the positions in a direction (Y direction) perpendicular to the grating direction (Z direction). Because each of the correction curves **20***a* represents an assignment relation between values that relate to X-ray absorption of the subject **90** and values that relate to X-ray scattering of the subject at corresponding one of the positions, a change degree of a spectrum of X-rays depending on an incident angle of the X-rays at each position is taken into account. Also, the first dark field image **35***a* can be accurately corrected by correcting the first dark field image **35***a*, which is a dark field image of a subject **90**, by using the correction curves **20***a* corresponding to the positions, as compared with a case in which the first dark field image **35***a* is corrected by using a single common correction curve **20***a*. Consequently, it is possible to reduce deterioration of image quality of a first dark field image **35***a* even in a case in which X-rays are incident on a grating in an inclined direction. Also, according to the X-ray image processing apparatus **6** and the X-ray image processing method according to the aforementioned embodiment, similar to the phase contrast X-ray imaging apparatus **100** according to the aforementioned embodiment, it is possible to reduce deterioration of image quality of a first dark field image **35***a* even in a case in which X-rays are incident on a grating in an inclined direction.

[0144] Also, in the aforementioned embodiment, the correction curve generation method includes acquiring a second dark field image **31**, which is a dark field image of a correction phantom **13**, by capturing an image of the correction phantom **13** using a phase contrast X-ray imaging apparatus **100** including a plurality of gratings; acquiring a second absorption image **32**, which is an absorption image of the correction phantom **13**; and generating the plurality of correction curves **20***a* at positions in the direction (Y direction) perpendicular to the grating direction (Z direction) in which the plurality of gratings extend by using the second dark field image **31** and the second absorption image **32**. Accordingly, because the plurality of correction curves **20***a* are generated by using the second dark field image **31** and the second absorption image **32** acquired by actually capturing the correction phantom **13**, it is possible to generate the plurality of correction curves **20***a* with a change degree of a spectrum of X-rays depending on an incident angle of the X-rays at each position being highly accurately being reflected.

[0145] In addition, following additional advantages can be obtained by the aforementioned embodiment added with configurations discussed below.

[0146] In other words, in this embodiment, as described above, the image processor **7***a* generates a first absorption image **35***b* of the subject **90**, which is an absorption image of the subject **90**, in addition to the first dark field image **35***a*; and the controller **7***b* is configured to access, for each of the positions, corresponding one of the correction curves **20***a*, to specify values that relate to the X-ray scattering and are assigned to values that relate to the X-ray absorption in the first absorption image **35***b*, and to correct the first dark field image **35***a* by using the specified values, which relate to the X-ray scattering. In addition, the controller **7***b* acquires values that relate to the X-ray absorption at the aforementioned positions in the first absorption image **35***b*. The controller **7***b* accesses, for each of the positions, corresponding one of the correction curves **20***a*, and acquires values that relate to X-ray scattering corresponding to the values that relate to X-ray absorption. Subsequently, the controller **7***b* corrects the first dark field image **35***a* by using the X-ray scattering values acquired. As a result, the first dark field image **35***a* can be accurately corrected by executing these processes as described above.

[0147] In this embodiment, as described above, the storage **8** is configured to store the plurality of correction curves **20***a* to associate the plurality of correction curves to materials of subjects **90**; and the controller **7***b* is configured to correct the first dark field image **35***a* by selecting, in accordance with a material of the subject **90**, corresponding ones of the plurality of correction curves **20***a*, and by using the selected ones of the plurality of correction curves **20***a*. Accordingly, because the correction curves **20***a* that correspond to the material of the subject **90** is selected, the first dark field image **35***a* can be corrected by the correction curve **20***a* that correspond to the X-ray absorptance of the subject **90**. Consequently, reduction of accuracy of correction of the first dark field image **35***a* is suppressed as compared with, for example, a configuration in which the first dark field image **35***a* is corrected by using one type of correction curve **20***a* independent of the material of the subject **90**, it is possible to further reduce deterioration of image quality of the first dark field image **35***a*.

[0148] In this embodiment, as described above, the storage **8** is configured to store the plurality of correction curves **20***a* to associate the plurality of correction curves to imaging conditions **22**; and the controller **7***b* is configured to correct the first dark field image **35***a* by selecting, in accordance with one of the imaging conditions **22**, corresponding ones of the plurality of correction curves **20***a*, and by using the selected ones of the plurality of correction curves **20***a*. Here, when setting of the imaging condition **22** is changed, a spectrum of X-rays detected by the X-ray detector **2** is changed. For this reason, in a case in which one type of correction curve **20***a* is used to correct the first dark field image **35***a* independent of the imaging condition **22**, accuracy of the correction may be reduced. To address this, the plurality of correction curves **20***a* that correspond to the imaging condition **22** set is used to correct the first dark field image **35***a* in the aforementioned configuration, and as a result it is possible to suppress reduction of correction accuracy of the first dark field image **35***a*.

Modified Embodiments

[0149] Note that the embodiment disclosed this time must be considered as illustrative in all points and not restrictive. The scope of the present invention is not shown by the above description of the embodiments but by the scope of claims for patent, and all modifications (modified embodiments) within the meaning and scope equivalent to the scope of claims for patent are further included.

[0150] For example, while the example in which the correction phantom **13** has a stepped shape in which its thickness in the X-ray irradiation direction (X direction) changes in steps in the Z direction has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, a thickness in an X-ray irradiation direction (X direction) of the correction phantom may continuously change as in a correction phantom **130** according to a first modified embodiment shown in FIG. **17**. Specifically, as shown in FIG. **17**, the correction phantom **130** according to the first modified embodiment may have an inclined surface **130***a*.

[0151] While the example in which the correction phantom **13** is integrally formed of a homogeneous material has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, the correction phantom **13** is not necessarily integrally formed. For example, the correction phantom may be constructed of a combination of a plurality types of plate-shaped materials as in a correction phantom **131** according to a second modified embodiment shown in FIG. **18**. Specifically, the correction phantom **131** according to the second modified embodiment is constructed of a combination of a first plate-shaped material **131***a*, a second plate-shaped material **131***b*, a third plate-shaped material **131***c*, and a fourth plate-shaped material **131***d*.

[0152] While the example in which the phase contrast X-ray imaging apparatus **100** captures an image with orientations of the gratings being fixed has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, the phase contrast X-ray imaging apparatus may include a grating rotation mechanism for rotating the plurality of gratings about an optical axis of X-rays, and may be configured to capture images while changing angles of the plurality of gratings at different.

[0153] When the plurality of gratings are rotated, an energy distribution of X-rays detected by the X-ray detector **2** changes. Difference between energy distributions of X-rays due to difference between different rotation angles of the plurality of gratings is now described with reference to FIGS. **19**(A) and **19**(B). Here, the difference between energy distributions of X-rays is represented by different hatching patterns indicated by legends **60** in illustrations of FIGS. **19**(A) and **19**(B). Specifically, the denser the hatching pattern (the narrower the distance between hatching lines), the lower the energy of the X-rays. Note that, although boundary lines are shown between areas of different X-ray energies, the energy of X-rays actually smoothly changes.

[0154] An illustration of FIG. **19**(A) shows an image **38***a* of an energy distribution of X-rays on the X-ray detector **2** when the plurality of gratings are orientated with their grating directions agreeing with the Z direction. In the illustration shown in FIG. **19**(A), because the grating directions agree with the Z direction, the energy distribution of the X-rays changes in the Y direction, which is perpendicular to the grating direction. Specifically, in the Y-direction, the energy of the X-rays in the center of the image **38***a* is low, and the energy of the X-rays increases toward the edge of the image **38***a*.

[0155] An illustration of FIG. **19**(B) shows an image **38***b* of an energy distribution of X-rays when the plurality of gratings are rotated about an axis of an irradiation axis direction of the X-rays (X direction). The image **38***b* shown in FIG. **19**(B) is an image of an energy distribution of X-rays when the plurality of gratings are rotated 45 degrees. When the plurality of gratings are rotated, the energy distribution of the X-rays correspondingly rotates in accordance with a direction of the X-rays. In other words, the image **38***b* is an image corresponding to the image **38***a* of the energy distribution that is rotated 45 degrees.

[0156] For this reason, even when the plurality of gratings are rotated about the irradiation axis of the X-rays, it is not necessary to capture a correction image of a correction phantom **13** (see FIG. **8**) that is rotated at the same angle as the gratings to use the correction image to correct the first dark field image **35***a* (see FIG. **14**(A)) because the first dark field image can be corrected by applying a coordinate transformation to the correction curves **20***a* previously stored in the storage **8** (see FIG. **1**). Accordingly, the controller **7***b* corrects the first dark field image **35***a* by converting coordinates of the plurality of correction curves **20***a*.

[0157] While the example in which the controller **7***b* corrects the first dark field image **35***a* by using the plurality of correction curves **20***a* that correspond to a material of the subject **90** in the plurality types of correction curve sets **20**, which are generated corresponding to materials of the subjects **90** has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, the controller may be configured to correct the first dark field image **35***a* by using the plurality of correction curves **20***a* included in one type of correction curve set independent of the material of the subject **90**. However, if the controller corrects the first dark field image **35***a* by using the plurality of correction curves **20***a* included in one type of correction curve set independent of the material of the subject **90**, accuracy of the correction decreases. For this reason, the controller is preferably configured to correct the first dark field image **35***a* by using the plurality of correction curves **20***a* included in one type of correction curve set.

[0158] Also, in the aforementioned embodiment, the phase contrast X-ray imaging apparatus **100** does not necessarily include the input acceptor **10**. That is, the controller **7***b* may automatically select the plurality of correction curves **20***a* by using the first dark field image **35***a* and the first absorption image **35***b*. For example, it can be conceived that techniques such as machine learning or pixel processing may be applied to the selection.

[0159] While the example in which the storage **8** stores the plurality of correction curve sets **20**, which are generated based on the correction images **33** captured under imaging conditions **22** different from each other, for the imaging conditions **22**, and the controller **7***b* corrects the first dark

field image **35***a* based on the plurality of correction curves **20***a* that correspond to the acquired imaging condition **22** has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, the storage **8** does not necessarily store the plurality of correction curve sets **20** for the imaging conditions **22**. In this case, the controller **7***b* can correct the first dark field image **35***a* by using one type of correction curve set **20** independent of the imaging condition **22**. However, if the storage **8** does not store the plurality of correction curve sets **20** for the imaging conditions **22**, accuracy of correction of the first dark field image **35***a* may decrease in a case in which the imaging condition **22** is changed. For this reason, the storage **8** preferably stores the plurality of correction curve sets **20** for the imaging conditions **22**.
[0160] While the example in which the X-ray image processing apparatus **6** generates the correction curves **20***a* has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, the X-ray image processing apparatus **6** may be configured to store the plurality of correction curves **20***a*, which are generated by another image processing apparatus.
[0161] While the example in which the X-ray image processing apparatus **6** generates the plurality of correction curves **20***a* by using the correction image **33** of the correction phantom **13**, which is captured with the correction phantom being orientated to make a direction in which a thickness of the correction phantom **13** agree with the grating direction (Z direction) has been shown in the aforementioned embodiment, the present invention is not limited to this. When the correction phantom **13** is placed, the direction in which the thickness of the correction phantom **13** changes does not necessarily agree with the grating direction. However, if the direction in which the thickness of the correction phantom **13** changes does not agree with the grating direction, it is difficult to compensate for uneven noise caused by X-rays incident on the grating in an inclined direction. For example, when the correction image **33** is captured, it is preferable to make the direction in which the thickness of the correction phantom **13** changes agree with the grating direction.
[0162] While the example in which when the controller **7***b* uses one type of correction curve set **20** to correct the first dark field image **35***a* has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, in a case in which the controller corrects a dark field image of a subject that includes different materials stacked in the illumination direction of X-rays, the controller may use a plurality of correction curve sets **20** to correct the dark field image. Specifically, in a case in which the controller corrects a subject that includes an acrylic material and an aluminum material stacked in the illumination direction of X-rays, the controller may use the correction curves **20***a* that are suitable for the acrylic materials and the correction curves **20***a* that are suitable for the aluminum material together. In the case in which the controller uses the correction curves **20***a* that are suitable for the acrylic materials and the correction curves **20***a* that are suitable for the aluminum material together, the controller can use the correction curves that are obtained by averaging the correction curves **20***a* that are suitable for the acrylic materials and the correction curves **20***a* that are suitable for the aluminum material to correct the dark field image.
[0163] While the example in which the phase contrast X-ray imaging apparatus **100** includes the X-ray image processing apparatus **6** has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, the phase contrast X-ray imaging apparatus **100** does not necessarily include the X-ray image processing apparatus **6**. The X-ray image processing apparatus **6** may be provided independently of the phase contrast X-ray imaging apparatus **100**. In this case, the first dark field image **35***a* can be corrected by processing the phase contrast X-ray image **30**, which is acquired by the phase contrast X-ray imaging apparatus **100**, by using the X-ray image processing apparatus **6**.
[0164] While the example in which the grating position adjustment mechanism **12** adjusts a position of the second grating **4** has been shown in the aforementioned embodiment, the present

invention is not limited to this. The grating position adjustment mechanism **12** may be configured to adjust a position of a grating other than the second grating **4**. The grating position adjustment mechanism **12** can adjust a position of any of the plurality of gratings.

[Modes]

[0165] The aforementioned exemplary embodiment will be understood as concrete examples of the following modes by those skilled in the art.

(Mode Item 1)

[0166] A phase contrast X-ray imaging apparatus includes an X-ray source; an X-ray detector for detecting the X-rays for irradiation from the X-ray source; a plurality of gratings arranged between the X-ray source and the X-ray detector; an image processor for generating a phase contrast X-ray image including a first dark field image, which is a dark field image of a subject, based on an intensity distribution of the X-rays detected by the X-ray detector; a storage for storing a plurality of correction curves generated at positions in a direction perpendicular to a grating direction in which the plurality of gratings extend; and a controller for correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions.

(Mode Item 2)

[0167] In the phase contrast X-ray imaging apparatus according to mode item 1, the image processor generates a first absorption image of the subject, which is an absorption image of the subject, in addition to the first dark field image; and the controller is configured to access, for each of the positions, corresponding one of the correction curves, to specify values that relate to the X-ray scattering and are assigned to values that relate to the X-ray absorption in the first absorption image, and to correct the first dark field image by using the specified values, which relate to the X-ray scattering.

(Mode Item 3)

[0168] In the phase contrast X-ray imaging apparatus according to mode item 1 or 2, the storage is configured to store the plurality of correction curves to associate the plurality of correction curves to materials of subjects; and the controller is configured to correct the first dark field image by selecting, in accordance with a material of the subject, corresponding ones of the plurality of correction curves, and by using the selected ones of the plurality of correction curves.

(Mode Item 4)

[0169] In the phase contrast X-ray imaging apparatus according to mode item 1 or 2, the storage is configured to store the plurality of correction curves to associate the plurality of correction curves to imaging conditions; and the controller is configured to correct the first dark field image by selecting, in accordance with one of the imaging conditions, corresponding ones of the plurality of correction curves, and by using the selected ones of the correction curves.

(Mode Item 5)

[0170] An X-ray image processing apparatus includes an image acquirer for acquiring a phase contrast X-ray image including a first dark field image, which is a dark field image of a subject; a storage for storing a plurality of correction curves generated at positions in a direction perpendicular to a grating direction in which a plurality of gratings extend; and a controller for correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions.

(Mode Item 6)

[0171] An X-ray image processing method includes acquiring a phase contrast X-ray image including a first dark field image, which is a dark field image of a subject; storing a plurality of

correction curves generated at positions in a direction perpendicular to a grating direction in which a plurality of gratings extend; and correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions.
(Mode Item 7)
[0172] A method for generating the plurality of correction curves to be used in the X-ray image processing method according to mode item 6, the correction curve generation method includes acquiring a second dark field image, which is a dark field image of a correction phantom, by capturing an image of the correction phantom by using a phase contrast X-ray imaging apparatus including a plurality of gratings; acquiring a second absorption image, which is an absorption image of the correction phantom; and generating the plurality of correction curves at positions in the direction perpendicular to the grating direction in which the plurality of gratings extend by using the second dark field image and the second absorption image.
(Mode Item 8)
[0173] In the correction curve generation method according to mode item 7, the correction phantom has a structure whose X-ray absorption degree changes in the grating direction of the plurality of gratings; and in the generating the plurality of correction curves, each of the plurality of correction curves is generated by acquiring an assignment relation between values that relate to the X-ray absorption and are specified based on the second absorption image, and values that relate to the X-ray scattering and are specified based on the second dark field image at corresponding one of the positions.
(Mode Item 9)
[0174] In the correction curve generation method according to mode item 7, the correction phantom has a structure whose thickness in an X-ray irradiation direction changes in the grating direction of the plurality of gratings; and in the generating the plurality of correction curves, each of the plurality of correction curves is generated by acquiring an assignment relation between values that relate to the X-ray absorption and are specified based on the second absorption image, and values that relate to the X-ray scattering and are specified based on the second dark field image at corresponding one of the positions.
DESCRIPTION OF REFERENCE NUMERALS
[0175] **1**; X-ray source [0176] **2**; X-ray detector [0177] **3**; first grating (one of gratings) [0178] **4**; second grating (one of gratings) [0179] **5**; third grating (one of gratings) [0180] **6**; X-ray image processing apparatus [0181] **7***a*; image processor [0182] **7***b*; controller [0183] **8**; storage [0184] **9**; image acquirer [0185] **13**, **13***e*, **13***f*, **130**, **131**; correction phantom [0186] **22**; imaging condition [0187] **30**; phase contrast X-ray image [0188] **31**; second dark field image (correction phantom dark field image) [0189] **32**; second absorption image (absorption image of correction phantom) [0190] **35***a*; first dark field image (dark field image of subject) [0191] **35***b*; first absorption image (absorption image of subject) [0192] **90**; subject [0193] **100**; phase contrast X-ray imaging apparatus

Claims

1. A phase contrast X-ray imaging apparatus comprising: an X-ray source; an X-ray detector for detecting the X-rays for irradiation from the X-ray source; a plurality of gratings arranged between the X-ray source and the X-ray detector; an image processor for generating a phase contrast X-ray image including a first dark field image, which is a dark field image of a subject, based on an intensity distribution of the X-rays detected by the X-ray detector; a storage for storing a plurality of correction curves generated at positions in a direction perpendicular to a grating direction in which the plurality of gratings extend; and a controller for correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions.

2. The phase contrast X-ray imaging apparatus according to claim 1, wherein the image processor generates a first absorption image of the subject, which is an absorption image of the subject, in addition to the first dark field image; and the controller is configured to access, for each of the positions, corresponding one of the correction curves, to specify values that relate to the X-ray scattering and are assigned to values that relate to the X-ray absorption in the first absorption image, and to correct the first dark field image by using the specified values, which relate to the X-ray scattering.

3. The phase contrast X-ray imaging apparatus according to claim 1, wherein the storage is configured to store the plurality of correction curves to associate the plurality of correction curves to materials of subjects; and the controller is configured to correct the first dark field image by selecting, in accordance with a material of the subject, corresponding ones of the plurality of correction curves, and by using the selected ones of the plurality of correction curves.

4. The phase contrast X-ray imaging apparatus according to claim 1, wherein the storage is configured to store the plurality of correction curves to associate the plurality of correction curves to imaging conditions; and the controller is configured to correct the first dark field image by selecting, in accordance with one of the imaging conditions, corresponding ones of the plurality of correction curves, and by using the selected ones of the plurality of correction curves.

5. An X-ray image processing apparatus comprising: an image acquirer for acquiring a phase contrast X-ray image including a first dark field image, which is a dark field image of a subject; a storage for storing a plurality of correction curves generated at positions in a direction perpendicular to a grating direction in which a plurality of gratings extend; and a controller for correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions.

6. An X-ray image processing method comprising: acquiring a phase contrast X-ray image including a first dark field image, which is a dark field image of a subject; storing a plurality of correction curves generated at positions in a direction perpendicular to a grating direction in which a plurality of gratings extend; and correcting the first dark field image by using the correction curves corresponding to the positions, wherein each of the plurality of correction curves represents an assignment relation between values that relate to X-ray absorption of the subject and values that relate to X-ray scattering of the subject at corresponding one of the positions.

7. A method for generating the plurality of correction curves to be used in the X-ray image processing method according to claim 6, the correction curve generation method comprising: acquiring a second dark field image, which is a dark field image of a correction phantom, by capturing an image of the correction phantom by using a phase contrast X-ray imaging apparatus including a plurality of gratings; acquiring a second absorption image, which is an absorption image of the correction phantom; and generating the plurality of correction curves at positions in the direction perpendicular to the grating direction in which the plurality of gratings extend by using the second dark field image and the second absorption image.

8. The correction curve generation method according to claim 7, wherein the correction phantom has a structure whose X-ray absorption degree changes in the grating direction of the plurality of gratings; and in the generating the plurality of correction curves, each of the plurality of correction curves is generated by acquiring an assignment relation between values that relate to the X-ray absorption and are specified based on the second absorption image, and values that relate to the X-ray scattering and are specified based on the second dark field image at corresponding one of the positions.

9. The correction curve generation method according to claim 7, wherein the correction phantom has a structure whose thickness in an X-ray irradiation direction changes in the grating direction of the plurality of gratings; and in the generating the plurality of correction curves, each of the plurality of correction curves is generated by acquiring an assignment relation between values that relate to the X-ray absorption and are specified based on the second absorption image, and values that relate to the X-ray scattering and are specified based on the second dark field image at corresponding one of the positions.